25900199 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCSTA100 - TIME SERIES METHODS AND THEIR APPLICATIONS

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **2** marks

1. What is the multiplicative model in time series analysis? [K] / [CO1]

   a). A model where components are multiplied

   b). A model where components are added together

   c). A model that assumes no trend

   d). A model that ignores seasonality

2. What is the main objective of time series analysis? [U] / [CO1]

   a). To find relationships between two variables

   b). To forecast future values based on past data

   c). To compare multiple datasets

   d). None of the above

3. What does stationarity in a time series imply? [K] / [CO1]

a). The mean and variance of the series change over time

b). The statistical properties remain constant over time

c). The series is always increasing

d). The time series contains no noise

4. Why is white noise assumed in residual analysis of time series models? [S] / [CO2]

a). To remove random fluctuations

b). To simplify trend detection

c). To confirm that no additional patterns remain in the data

d). To make seasonality analysis unnecessary

5. How does a random walk process differ from an AR process? [S] / [CO2]

a). A random walk depends only on the previous value, while an AR process depends on multiple past values

b). A random walk has a fixed mean, but an AR process does not

c). A random walk has periodic fluctuations

d). An AR process is always non-stationary

6. What type of correlation does an upward trend in a scatter plot indicate? [I] / [CO3]

a). Positive correlation b). Negative correlation

c). Weak correlation d). Zero correlation

7. If the PACF has a significant spike at lag 1, it suggests: [I] / [CO1, CO3]

a). An AR(1) process b). An MA(1) process

c). No autocorrelation d). A seasonal component

8. Which of the following is a possible cause of autocorrelation in time series data? [I] / [CO1, CO3]

a). Seasonality b). Trend

c). Measurement error d). All of the above

9. What does "AR" stand for in AR models? [S] / [CO1, CO2]

a). Autoregressive b). Auto-related

c). Auto-residual d). Autocorrelated

10. The built-in R function to plot the PACF is, [I] / [CO1, CO3]

a). Pacf() b). PACF() c). pacf() d). None of the above

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

11. Write a note on the difference between trend and seasonality of time series data? [An] / [CO1]

12. Differentiate stationary time series from non- stationary time series. [An] / [CO1]

13. Distinguish between month plot and season plot in time series analysis. [An] / [CO1]

14. How will you differentiate correlation from co-variance? [I] / [CO1, CO3]

15. Briefly explain different types of stationary processes. [S] / [CO2]

[5x3 = 15]